25901110

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBTG201, MG4DSCBBT202- IMMUNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. List the immune cells that present antigens to T cells [K] / [CO1]
   List the names of APCs- 2 mark (APCs- 1 mark)
2. Recall the organ that is specialized in filtering blood. [K] / [CO1]
   Spleen -2 mark
3. Name the type of transplantation between two genetically different individuals of the same species.[K] / [CO3]
   Allograft- 2 marks.
4. Name the immunological effect where the excess antibodies in a sample cause false-negative results in agglutination test.[K] / [CO2]
   Prozone phenomenon- 2 marks
5. Recall the immunoassay suitable for measuring drug levels in blood. [U] / [CO2]
   ELISA and RIA- 2 marks
6. Name the determinant that define the isotype of the antibody based on heavy chain present.[K] / [CO1]
   Isotypic determinant- 2 marks
7. Name the immunity transferred via placenta. [K] / [CO4]
   Natural passive immunity- 2 marks
8. Name one disease caused by Type III hypersensitivity. [U] / [CO3]
   Anyone- 2 marks
9. Identify the technique that separates and detects proteins using antibodies. [U] / [CO2]
   Western Blotting- 2 marks
10. List any two examples of molecular mimicry. [K] / [CO3]
    Any two examples- 2 marks
11. Select the immunity that lacks memory [K] / [CO1]

Innate immunity- 2 marks- 2 marks

12. Name the antigens which originate from own body of the host organisms. [K] / [CO1]

Auto antigens / self-antigens- 2 marks [2x10 = 20]

Part B-Short Essay-Answer any 5 questions

Each question carries 4 marks

13. Describe the procedure of CFT step by step. [U] / [CO2]

Definition- 1 mark, steps (or figure with explanation)- 3 marks

14. Outline the structural components of MHC Class II molecules with labelled diagram.[K] / [CO1]

MHC- definition- 1 mark, structural components of MHC Class II molecules- 2 marks, diagram- 1 mark.

15. Differentiate congenital and acquired immunodeficiency diseases. [U] / [CO3]

Immunodeficiency diseases-introduction- 1 mark, congenital immunodeficiency diseases- 1.5 marks, acquired immunodeficiency diseases-1.5 marks

16. List the features of active and passive immunity [K] / [CO1]

Active immunity- features (4 points)- 2 marks, passive immunity- features(4 points)- 2 marks

17. Explain the steps involved in radioimmunodiffusion. [U] / [CO2]

Radioimmunodiffusion -introduction- 1 mark, steps- 3 marks.

18. Recall the characteristics of hybridoma cells [U] / [CO4]

Hybridoma cells- introduction-2 marks, characteristics of hybridoma cells(minimum 2 points)- 2 marks

19. Explain the applications of Widal test. [U] / [CO2]

Introduction- 2 marks, Applications (any 2 points)-2 marks [4x5 = 20]

Part C

Essay type questions-Answer any 1 questions

Each question carries 10 marks

20. Explain recombinant antibody production and explain its advantages. [U] / [CO4]

Recombinant antibody production- introduction-2 marks, Production- steps- 5 marks, Advantages (3 points) - 3 marks

21. Show the cells involved, phases, and effector mechanisms of cell-mediated immune response.[K] / [CO1]

Cells involved- 4 marks, phases- activation, maturation- 4 marks, effector mechanisms-2 marks.

[10x1 = 10]